AUTONOMOUS VOICE TELEPHONY & IN-CALL COMMERCE INFRASTRUCTURE

Enterprise Voice Operations & In-Call Transaction System.

A unified, sub-100ms voice operations platform engineered for Indian enterprises. Executes real-time dynamic customer interactions, synchronizes business catalogs, generates GST invoices with direct UPI settlement during live calls, and enforces carrier-grade regulatory compliance.

01 • TELEPHONY CORE <100ms

Native SIP/PSTN interconnect with Indian carriers (Airtel, Jio, Vi). Natural conversation pace with zero robotic start-up delay and adaptive ambient noise cancellation.

02 • OPERATIONS BRAIN Live Sync

Dynamic enterprise memory updated in real time verbally or via API. Instantly reflects doctor/service slots, inventory changes, and special commercial pricing without code pushes.

03 • IN-CALL COMMERCE Direct UPI

Automated statutory GST invoice generation dispatched to WhatsApp while caller is on the line. Non-custodial settlement routed directly to client merchant bank accounts.

04 • VISION VERIFICATION 12-Digit UTR

Computer vision parsing of incoming WhatsApp payment screenshots across Google Pay, PhonePe, and Paytm with automated ledger reconciliation and owner alerts.

Enterprise System Topology

Vaakriti unifies carrier telephony, conversational intelligence, dynamic enterprise catalog retrieval, and digital payment infrastructure into an integrated, zero-loss transaction engine.

END-TO-END AUTONOMOUS TRANSACTION FLOW



CORE PLATFORM SUBSYSTEMS

SUBSYSTEM	OPERATIONAL SCOPE	DELIVERY GUARANTEE
Voice Runtime Engine	Full duplex acoustic processing, intelligent interruption handling, sub-100ms end-to-end turnaround.	Zero artificial pauses, conversational cadence
Dynamic Knowledge Store	Real-time synchronization of medical appointment slots, retail SKU inventories, and pricing tables.	Instant write/read propagation (<500ms)
In-Call Invoicing Gateway	Dynamic PDF generation with SAC/HSN codes, verified GST calculations, and WhatsApp delivery during the call.	100% compliant tax documentation
Warm Telephony Failover	Dual-SIM call forward loop interception, automated VIP caller routing, and 20s spoken reassurance intervals.	Anti-voicemail loop prevention
Vision OCR Reconciler	Automated extraction of 12-digit bank UTR numbers and receipt verification from GPay, PhonePe, and Paytm.	Zero manual screenshot checking

The Operations Brain

Traditional voice bots rely on static FAQs that quickly become obsolete. The Vaakriti Operations Brain maintains continuous, bi-directional synchronization with your daily operational realities.

PILLAR 01

Spoken Voice Ingestion

Business owners and clinic coordinators update business rules by speaking naturally to their dashboard. No spreadsheets or technical updates required.

> *"Doctor Sharma 4 PM se pehle available nahi hain."*

PILLAR 02

Real-Time Availability

Direct calendar and appointment ledger binding ensures the AI never offers booked time slots or promises out-of-stock inventory to callers.

> *"Only 3 consultation slots remaining for today."*

PILLAR 03

Dynamic Price Policies

Instantaneous adjustment of emergency charges, diagnostic discounts, or delivery fees applied automatically across all concurrent calls.

> *"Apply festive 10% discount on health checkups."*

OPERATIONAL SPECIFICATIONS & DATA INGESTION

- **Atomic Ledger Updates:** Operations updates take effect globally across active and future SIP channels within <500ms.
- **Multi-Channel Ingestion:** Accepts updates via WhatsApp voice notes, dashboard web forms, or REST Webhook APIs.
- **Automated Conflict Resolution:** Flags overlapping appointments and alerts the business owner before confirming with caller.
- **Tamper-Proof Audit Logging:** Full historical ledger of all knowledge modifications with user identity and timestamps.

Real-Time In-Call Invoicing

VAAKRITI

Eliminates the typical 85% drop-off* that occurs when phone leads are sent payment links hours after a call. Vaakriti issues compliant GST invoices and dynamic UPI QR codes during the active call.

IN-CALL TRANSACTION LIFECYCLE

1. Verbal Agreement — T = 0S

Caller agrees to service booking, consultation fee, or product order. AI confirms customer name and billing details.

2. PDF Invoice Assembly — < 1.2S

System formats official GST invoice with sequential numbering, SAC/HSN codes, CGST/SGST/IGST breakdown, and business branding.

3. WhatsApp Delivery While on Call — LIVE CALL

Invoice PDF and dynamic UPI QR code are delivered directly to the caller's WhatsApp number before the call disconnects.

4. Non-Custodial Direct Settlement — RBI COMPLIANT

Payment settles 100% directly into the business's registered bank account (HDFC, ICICI, SBI) via UPI Intent/VPA. Zero intermediary wallet holding.

TAX INVOICE

GSTIN: 23RAGHAVTHAKUR000

INV # VAAK/26-27/0042
DATE: 04/11/2026

Billed To: Rahul Sharma (+91 98765 43210)
Origin: Autonomous AI Phone Booking

Description	HSN	Amount
Executive Health Consultation	998311	**₹1,200.00**
	Taxable Subtotal:	₹1,200.00
	GST (18%):	₹216.00
	Grand Total:	**₹1,416.00**

UPI INSTANT PAY ENABLED
VPA: vaakriti@hdfc • Dynamic QR code embedded in WhatsApp dispatch

*Based on internal estimates; results vary by deployment.

Receipt OCR & Fail-Safe Transfer

VAAKRITI

Complete operational closure: automatically verifies customer payment screenshots submitted on WhatsApp and provides carrier-grade warm call transfer with zero infinite loops.

OCR VERIFICATION MULTI-APP PARSER

WhatsApp Payment Audit

When a customer transfers funds via UPI and sends a screenshot to the WhatsApp line, the vision engine automatically extracts and cross-checks critical settlement metadata.

EXTRACTED FIELD	VALIDATION PROTOCOL
12-Digit Bank UTR	Regex pattern matched & checked against historical database to eliminate duplicate receipt replay attacks.
Transaction Status	Distinguishes between "Success", "Pending", and "Failed" indicators across GPay, PhonePe, and Paytm.
Settlement Amount	Matched exactly against expected pending GST invoice grand total before issuing official receipt.
Instant Alert	Dispatches immediate confirmation to customer and notifies business owner via dashboard push notification.

TELEPHONY ESCALATION ANTI-LOOP ARCHITECTURE

Loop-Proof Warm Transfer

Solves the catastrophic "forwarding loop" where a call forwarded to an owner's busy phone rings back into the bot, trapping customers in infinite automated loops.

SAFETY MECHANISM	ENTERPRISE SAFEGUARD
Dual-SIM Routing	Routes transfers exclusively to designated secondary operational numbers, bypassing primary forward routes.
Loop Interception	Detects carrier voicemail tones and busy signals instantly; reclaims the active channel instead of dropping call.
VIP Caller Whitelist	Recognized VIP clients, family, and enterprise directors are routed directly to the owner without bot friction.
Caller Reassurance	Plays contextual spoken status updates every 20 seconds during transfer wait times, preventing drop-offs.

Carrier Connectivity & Audio Quality

Built specifically for Indian telecom realities. Engineered to eliminate latency, handle noisy real-world backgrounds (traffic, market chatter), and deliver crisp, natural voice exchanges.

Sub-100ms Turnaround — TARGET LATENCY

Optimized pipeline with dedicated streaming audio chunking. Eliminates unnatural 2-3 second robotic lag common in generic cloud voice bots.

Indian Ambient Filtering — 8KHZ / 16KHZ

Adaptive neural noise suppression tuned for Indian background acoustics: vehicle horns, ceiling fans, street markets, and low-bandwidth 2G/4G coverage.

Hindi & Hinglish Fluency — NATIVE ACOUSTIC

Native comprehension of code-mixed Hindi-English sentences, regional inflections, colloquial terms, and Indian numerical formats (Lakhs, Crores).

Carrier Interconnects — SIP / PSTN

Direct SIP trunking and cloud PSTN routing across Bharti Airtel, Reliance Jio, Vodafone Idea, Tata Tele Business Services, Exotel, and Twilio.

AUDIO LATENCY & NETWORK BENCHMARKING

PERFORMANCE METRIC	GENERIC CLOUD BOT	TRADITIONAL IVR	VAAKRITI ENTERPRISE
Response Latency	1,800ms - 3,200ms	Keypress Driven (Manual)	**< 100ms Streaming**
Voice Interruption (Barge-In)	Clipped / Unstable	Unsupported	**Instant Full-Duplex**
Audio Jitter Management	Noticeable crackles	Static circuit	**Adaptive Anti-Jitter Buffer**
Hinglish Code-Mixing	Frequent misunderstandings	Restricted to numbers	**Native Colloquial Accuracy**

Enterprise Security & TRAI Standards

Engineered with non-negotiable security guardrails (SEC-001 through SEC-011) ensuring complete compliance with Indian telecommunications, privacy, and RBI digital payment standards.

TRAI & DLT Calling Governance

Outbound calls are restricted to lawful calling windows (09:00 to 21:00 IST). Automated scrubbing against the National Do Not Disturb (DND) registry ensures full immunity from regulatory penalties.

Mandatory AI Identity & Recording Disclosure

Every single call begins with mandatory disclosures ("Main Trinetra/Vaakriti se Arika bol rahi hoon, ek AI assistant... Service quality ke liye yeh call record ki ja sakti hai") safeguarding enterprise transparency.

RBI Non-Custodial Direct Settlement

Vaakriti never holds customer money. Payments execute directly from caller UPI apps into your designated corporate current account, fully conforming with RBI merchant settlement norms.

Tamper-Evident SHA-256 Audit Trail

Every conversation transcript, consent verification, and financial reconciliation is written to an immutable audit register secured by row-level cryptographic signatures.

ENTERPRISE DATA PROTECTION STANDARDS

Tenant Isolation: Strict multi-tenant schema partitioning ensuring zero cross-tenant data leakage.

Transport Security: End-to-end TLS 1.3 encryption across all SIP signaling, WebSockets, and REST APIs.

Key Governance: Zero client-side exposure of service keys; tokenized temporary sessions only.

VAAKRITI

4-Day Production Go-Live

Rapid, zero-friction enterprise deployment lifecycle executed by dedicated solutions architects.

DAY 01

Knowledge Ingestion

Catalog import, consultation schedules, clinic services, GST details, and corporate brand tone definition.

DAY 02

Telephony Interconnect

SIP trunk provisioning, virtual business numbers, carrier porting, and WhatsApp Business API integration.

DAY 03

Payment & Smoke Tests

UPI gateway binding, live test transactions, receipt OCR verification, and simulated carrier network calls.

DAY 04

Production Go-Live

Live traffic cutover, operational dashboard access for staff, and 24/7 dedicated telemetry monitoring.

ENTERPRISE SERVICE LEVEL COMMITMENTS

99.9% Telephony Uptime
Carrier redundancy across primary Airtel and secondary Jio SIP backbones.

< 15 Min Critical Incident Response
Direct access to dedicated Tier-3 voice infrastructure engineering team.

Continuous Brain Updates
Unrestricted operational updates via dashboard or spoken voice interface.

ENTERPRISE ENGAGEMENT DESK

Schedule Your Infrastructure Assessment

Consult with our telephony and solutions architecture team for custom enterprise rollouts.

DIRECT ENTERPRISE LINE

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